BLUESTOCK FINTECH

Mutual Fund Analytics

Capstone Project Report

Intern: Manish | June 2026

Data Engineering | Financial Analytics | Power BI Dashboard

10 Datasets • 40 Fund Schemes • 32,778 Transactions

7 Days • 15+ Charts • 4-Page Dashboard • SQLite Star Schema

1. Executive Summary

This report documents a complete end-to-end mutual fund analytics project built during a 7-day internship at Bluestock Fintech. The project ingests 10 real-world datasets covering 40 mutual fund schemes, 32,778 investor transactions, 4 years of daily NAV history, and quarterly AUM figures across 10 Indian Asset Management Companies.

The pipeline transforms raw CSV data into a cleaned SQLite star schema, computes quantitative risk metrics including Value at Risk, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown, and delivers an interactive 4-page Power BI dashboard alongside Jupyter-based analytical notebooks.

Key Headline Numbers

Metric	Value	Source
Total Industry AUM (2025)	Rs.81 Lakh Crore	AMFI quarterly data
Monthly SIP ATH (Dec 2025)	Rs.31,002 Crore	AMFI monthly stats
Total Investor Folios	26.12 Crore	Industry folio data
NAV Records Processed	64,320 rows	After weekend forward-fill
Investor Transactions	32,778 records	5,000 unique investors
Highest 3-Yr Fund Return	23.4% CAGR	SBI Small Cap Direct
Best Sharpe Ratio	1.09	HDFC Top 100 Direct
Best Annualised Alpha	+27.2%	ABSL Small Cap Direct

2. Data Sources

The project uses 10 structured CSV datasets provided by Bluestock Fintech, supplemented by live NAV data fetched from the public mfapi.in REST API. All datasets were validated, cleaned, and loaded into a SQLite database before analysis.

#	File	Rows	Key Columns
1	01_fund_master.csv	40	amfi_code, fund_house, category, risk_category
2	02_nav_history.csv	64,320	amfi_code, date, nav (after forward-fill)
3	03_aum_by_fund_house.csv	90	date, fund_house, aum_crore
4	04_monthly_sip_inflows.csv	48	month, sip_inflow_crore, yoy_growth_pct
5	05_category_inflows.csv	144	month, category, net_inflow_crore
6	06_industry_folio_count.csv	21	month, total_folios_crore
7	07_scheme_performance.csv	40	amfi_code, sharpe_ratio, alpha, beta
8	08_investor_transactions.csv	32,778	investor_id, transaction_type, amount_inr
9	09_portfolio_holdings.csv	322	amfi_code, sector, weight_pct
10	10_benchmark_indices.csv	8,050	date, index_name, close_value

Live NAV API

Live and historical NAV data was fetched from mfapi.in using HTTP GET requests to https://api.mfapi.in/mf/{amfi_code}. The API returns JSON with fund metadata and all daily NAV records. Six key schemes were fetched yielding 19,798 additional NAV records saved to data/raw/.

3. ETL Design

3.1 Extraction

Raw CSVs are loaded using pandas read_csv() with explicit dtype inference. The data_ingestion.py script prints shape, dtypes, and head(3) for each dataset and validates AMFI code cross-references between fund_master and nav_history — confirming all 40 codes match with zero orphans.

3.2 Transformation

• nav_history: dates parsed to datetime, sorted by amfi_code + date, forward-filled weekends and holidays expanding 46,000 to 64,320 rows, duplicates removed, NAV greater than 0 validated.

• investor_transactions: SIP/Lumpsum/Redemption labels standardised, all amounts greater than 0 validated, KYC status enum values checked — only Verified and Pending allowed.

• scheme_performance: all return columns validated as numeric, expense_ratio range checked from 0.55% to 1.64%, within SEBI limit of 0.1% to 2.5%, anomalies flagged.

• All other datasets: date parsing to YYYY-MM-DD string format, numeric type coercion, duplicate row removal.

3.3 Loading

SQLAlchemy create_engine() with SQLite backend is used. All 10 cleaned datasets are loaded via df.to_sql(). The star schema includes 2 dimension tables and 5 fact tables plus 4 supplementary tables. Row counts are verified dynamically against source DataFrames. A data quality report is saved automatically to reports/.

Table	Rows	Type
dim_fund	40	Dimension
dim_date	1,608	Dimension
fact_nav	64,320	Fact
fact_transactions	32,778	Fact
fact_performance	40	Fact
fact_aum	90	Fact
fact_portfolio	322	Fact
sip_inflows	48	Supplementary
category_inflows	144	Supplementary
folio_count	21	Supplementary
benchmark_indices	8,050	Supplementary

4. EDA Findings

4.1 Industry Growth

The Indian mutual fund industry has seen extraordinary growth between 2022 and 2025. Monthly SIP inflows grew 169% from Rs.11,517 Crore in January 2022 to an all-time high of Rs.31,002 Crore in December 2025. Total industry folios nearly doubled from 13.26 Crore to 26.12 Crore, reflecting massive retail investor onboarding driven by digital platforms.

4.2 Fund House Rankings

SBI Mutual Fund leads all AMCs with Rs.12.5 Lakh Crore AUM as of 2025 — nearly double its 2022 position of Rs.6.05 Lakh Crore. ICICI Prudential at Rs.9.2 Lakh Crore and HDFC at Rs.8.8 Lakh Crore round out the top three. DSP Mutual Fund and Mirae Asset remain the smallest in the dataset.

4.3 Investor Demographics

- The 26-35 age group accounts for 41% of all transactions — digitally native investors comfortable with SIPs and mobile investing.
- Males outnumber female investors approximately 2 to 1, indicating room for growth in female financial inclusion.
- Punjab and Tamil Nadu lead state-wise SIP investments despite being geographically distant.
- T30 cities account for 66% of transactions while B30 cities contribute 34% — growing through digital onboarding.
- UPI has become the dominant payment mode, overtaking Mandate and Cheque for SIP transactions.

4.4 Category Analysis

Small Cap and Mid Cap categories consistently attract the highest net inflows. Liquid funds show negative net inflows in certain months, indicating short-term redemption pressure. Large and Mid Cap funds show the most consistent positive inflows across all months in the category heatmap analysis.

5. Performance Analysis

5.1 Return Analysis (CAGR)

CAGR was computed from raw NAV data using the formula: (NAV_end / NAV_start)^(1/n) minus 1. Since data starts from January 2022, we compute 1-year, 3-year, and 4-year CAGR for all 40 schemes.

Fund (Direct Plan)	1-Yr CAGR	3-Yr CAGR	Category
SBI Small Cap Direct	~24.9%	23.4%	Small Cap
ABSL Small Cap Direct	~24.9%	22.4%	Small Cap
Axis Midcap Direct	~21.9%	18.5%	Mid Cap
HDFC Top 100 Direct	~15.2%	16.2%	Large Cap
ICICI Pru Bluechip Direct	~13.9%	14.4%	Large Cap

5.2 Risk Metrics

• Sharpe Ratio: HDFC Top 100 Direct leads at 1.09, followed by SBI Small Cap Direct at 1.07. Liquid and debt funds show Sharpe of 5 to 7 due to near-zero volatility.

• Alpha: SBI Small Cap Direct shows annualised alpha of +27.2% above NIFTY 100 — strong fund manager value-add beyond market exposure.

• Beta: Most Large Cap funds have beta between 0.85 and 0.95 — slightly defensive relative to NIFTY 100.

• Max Drawdown: DSP Small Cap at -35.4% and SBI Small Cap at -28.1% showed worst peak-to-trough declines during the 2024 correction.

• VaR 95%: Liquid funds show -0.08% to -0.15%. Small Cap funds show VaR of -3% to -4% at 95% confidence.

5.3 Fund Scorecard

A composite score of 0 to 100 was computed with: 30% weight to 3-year CAGR rank, 25% Sharpe rank, 20% Alpha rank, 15% Expense ratio rank (inverse), and 10% Max Drawdown rank (inverse). SBI Small Cap Direct topped at 92.4, followed by HDFC Top 100 Direct at 88.1.

6. Power BI Dashboard

The interactive dashboard was built in Power BI Desktop with 4 pages and cross-filtering enabled across all visuals. A dark navy and teal colour theme was applied consistently. All 8 cleaned CSV tables were imported and relationships built on amfi_code and date keys.

Page 1 — Industry Overview

4 KPI cards for Total AUM, SIP ATH, Folios, and Schemes count. AUM growth chart by year. AUM by fund house bar chart showing SBI dominance. A DAX measure was written for correct latest-quarter AUM calculation.

Page 2 — Fund Performance

Scatter plot of 3-year return versus standard deviation with bubble size as AUM. Sortable fund scorecard table with returns, Sharpe ratio, and expense ratio. Three interactive slicers for fund house, category, and plan.

Page 3 — Investor Analytics

Transaction amount by state horizontal bar chart. SIP, Lumpsum, and Redemption donut chart. Age group versus average SIP amount column chart. City tier pie chart. Slicers for state, age group, and city tier.

Page 4 — SIP and Market Trends

Dual-axis chart with SIP inflow bars and SIP AUM trend line. Category inflow matrix heatmap with conditional colour formatting. Top categories by net inflow bar chart.

7. Advanced Analytics

7.1 Value at Risk and CVaR

Historical VaR at 95% confidence was computed as the 5th percentile of each fund's daily return distribution. CVaR is the mean return of the worst 5% of days. Small Cap funds show VaR of -3% to -4% while Liquid funds show -0.08% to -0.15%.

7.2 Rolling 90-Day Sharpe

Rolling Sharpe was computed for 5 key funds. The chart reveals that all funds dipped sharply in late 2024 during the market correction, with SBI Small Cap showing the most volatile rolling Sharpe — swinging from above 2.0 in 2023 to near 0 during the correction period.

7.3 Investor Cohort Analysis

Investors grouped by first transaction year. The 2024 cohort of 4,803 investors invested Rs.2,500 Crore total at an average of Rs.1.08 Lakh per transaction. The 2025 cohort of 4,018 investors shows similar average tickets, suggesting consistent investor quality across cohorts.

7.4 SIP Continuity

Among 1,362 investors with 6 or more SIP transactions, the average gap between installments was computed. Investors with gaps exceeding 35 days were flagged as at-risk. The 17-month data window means regular monthly investors can show average gaps slightly above 30 days — genuine at-risk investors show significantly higher gaps.

7.5 Sector HHI Concentration

The Herfindahl-Hirschman Index (HHI equals the sum of sector weight squared) was computed for all equity funds. Index and ETF funds showed the highest HHI (most concentrated), while Flexi Cap funds showed the lowest HHI, confirming better sector diversification.

8. Limitations

• Data period starts from January 2022. Some funds have limited 5-year history, reducing CAGR reliability for newer schemes.

• Investor transaction data is synthetic and anonymised — real-world investor behaviour may differ from observed patterns.

• The fund recommender is rule-based, not ML-based. It does not account for correlation between recommended funds.

• Power BI dashboard requires Power BI Desktop. A web-published version would improve accessibility.

• NIFTY 100 was used as a single benchmark for all alpha/beta regression. Category-specific benchmarks would be more accurate.

• The composite scorecard weights are subjective — different weightings would change fund rankings.

• Portfolio holdings data has a single date of 2025-12-31, preventing temporal portfolio evolution analysis.

9. Recommendations

For Investors

• Prefer Direct plans over Regular plans — the expense savings compound significantly over long investment horizons.

• Add Small Cap or Debt funds to portfolios dominated by Large Cap funds to reduce the high 0.92 to 0.97 correlation risk.

• Focus on Sharpe Ratio and Max Drawdown when comparing funds — not just raw returns in isolation.

• Maintain SIP continuity — even small gaps reduce the rupee-cost-averaging benefit of systematic investing.

For Bluestock Fintech

• Publish the Power BI dashboard to Power BI Service for browser access without requiring Desktop installation.

• Add a machine learning layer to the fund recommender using collaborative filtering on investor transaction history.

• Expand the investor dataset to 3 or more years of history for richer cohort and retention analysis.

• Build real-time NAV alerts using mfapi.in to flag when a fund crosses its 52-week high or low.

• Develop a SIP churn prediction model using the 35-day gap threshold as a training label.

10. Self-Review Checklist

Deliverable	Status	Notes
All 10 CSV datasets loaded and explored	Done	data_ingestion.py
Live NAV fetched from mfapi.in	Done	live_nav_fetch.py
All 10 datasets cleaned	Done	data/processed/
SQLite star schema designed	Done	sql/schema.sql
All data loaded via SQLAlchemy	Done	db_loader.py
10 analytical SQL queries written	Done	sql/queries.sql
Data dictionary documented	Done	reports/data_dictionary.md
15+ EDA charts created	Done	EDA_Analysis.ipynb
Performance metrics computed	Done	Performance_Analytics.ipynb
Power BI dashboard — 4 pages	Done	dashboard/
VaR, CVaR, Rolling Sharpe computed	Done	Advanced_Analytics.ipynb
Fund recommender built and tested	Done	recommender.py
Sector HHI analysis complete	Done	Advanced_Analytics.ipynb
Final report written	Done	reports/Final_Report.pdf
12-slide presentation created	Done	Bluestock_MF_Presentation.pptx
README.md written	Done	README.md
GitHub repo tagged v1.0	Done	git tag v1.0